

Minimum Wage Many-Analyst Experiment

Executive Summary and Project Details

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1 Executive summary

This project tests whether **prompted literature priors** causally change what autonomous AI agents report about the employment effects of the minimum wage. Agents analyze the **same** state-year panel with the **same** task instructions; only the literature context (if any) is randomized. Multiple AI models are tested to examine whether model choice mediates the influence of literature primes.

Design. Five arms $\times$ 50 Cursor agents per model. Arms: no prime; null-effect short; null-effect with citations; negative-effect short; negative-effect with citations. Agents choose outcome, sample window, treatment definition, covariates, and estimator under a shared DiD methodology guide.

Headline results — Composer-2 run (Second Step, 250 agents).

Finding	Result
Completion	250/250 agents returned usable results.csv
Between-arm ATT differences	Not significant (Fisher omnibus $p = 0.4101$; Kruskal-Wallis $p = 0.3063$)
Outcome choice	~99.6% studied teen employment-to-population
Estimator choice	~98.8% Callaway-Sant’Anna family; 4% TWFE in any arm
Treatment definition	249/250 binary ; 0% continuous MW treatment
Short vs citation-rich primes	No meaningful ATT gap on planned contrasts

Headline results — Opus 4.6 run (Third Step, 249 agents, 1 failure).

Finding	Result
Completion	249/250 agents returned usable results.csv
Between-arm ATT differences	Significant (Fisher omnibus $p = 0.0010$; Kruskal-Wallis $p = 0.0073$)
Outlier arm	negative_short : mean ATT = -0.0027, median = -0.0085 (most negative)
Specification variation	negative_short shows 3 continuous treatment , 2 CGBS , 1 TWFE — more diverse than other arms
Pairwise	null_cites vs negative_short $p = \text{NA}$

Headline results — Grok 4.6 v1 (with CS recipe, 245 agents).

Finding	Result
Between-arm ATT differences	Highly significant (Fisher $p = 0.0000$; KW $p = 7.0\text{e-}12$)
Planned contrasts	Null Short – Neg Short $p < 0.0001$; Null Cites – Neg Cites $p = 0.0004$
Mechanism	Both negative arms shift left via outcome variable + control group choice (100% CS, 100% binary)

Headline results — Grok 4.6 v2 (neutralized instructions, 243 agents).

Finding	Result
Between-arm ATT differences	Highly significant (Fisher $p = 0.0000$; KW $p = 9.3e-07$)
Estimator adoption	100% Callaway–Sant’Anna despite removing CS recipe
Implication	CS preference is from model training, not the instruction document

Bottom line. Literature primes produce **model-dependent** effects through **different specification channels**:

- **Composer-2:** No detectable prime effect (Wave 1-like).
- **Opus 4.6: negative_short** arm diverges via estimator/treatment switching (Wave 2-like).
- **Grok 4.6:** Both negative arms diverge via outcome variable and control group selection — **the strongest prime effect observed**, and the mechanism persists identically when the CS step-by-step recipe is removed.

Implication. The mechanism through which priors affect estimates varies by model. Removing instructional scaffolding does not eliminate the prime effect — agents’ specification choices are shaped by model training, not the methodology document.

2 1. Purpose and motivation

Many-analyst studies show that researchers facing identical data can report different causal estimates through **specification choices**. Borjas & Breznau (2026, *Science Advances*) linked immigration researchers’ priors to their estimates. Scott Cunningham’s design turns that association into an **experiment** by randomizing the prior via a literature prime while holding data and task fixed ([Substack write-up](#)).

This repository is a **Cursor SDK replication / extension** of Cunningham’s `mw-analyst-experiment` design (originally Claude Code CLI).

Research question: Do literature primes about minimum-wage employment effects causally shift autonomous agents’ reported average treatment effects (ATTs)?

3 2. Experimental design

3.1 Arms

Arm ID	Label	Literature file
<code>none</code>	None	<i>(none)</i>
<code>null_short</code>	Null Short	<code>PRIME_NULL_SHORT.md</code>
<code>null_cites</code>	Null Cites	<code>PRIME_NULL_CITES.md</code>
<code>negative_short</code>	Negative Short	<code>PRIME_NEGATIVE_SHORT.md</code>
<code>negative_cites</code>	Negative Cites	<code>PRIME_NEGATIVE_CITES.md</code>

All agents receive: `agent_panel_essential.csv`, `DATA_DICTIONARY.md`, `DID_METHODODOLOGY.md`, and `INSTRUCTIONS_SHARED.md`. Primed arms also receive one literature context file.

3.2 What agents decide

- Employment outcome (CPS subgroups, QCEW industries, etc.)
- Treatment definition (binary staggered vs continuous)
- Sample years / states
- Covariates and comparison group
- Estimator (Callaway–Sant’Anna, continuous DiD, TWFE, etc.)

3.3 Deliverables per agent

1. **results.csv** — one row with design choices and ATT / SE / CI
2. **llms.txt** — structured narrative of the analysis

3.4 Data

Pre-aggregated **state** × **year** panel (51 units, 1990–2022): effective minimum wages, CPS employment rates for multiple groups, and selected QCEW industry series.

3.5 Scale and waves

Wave	Location	Scale	Model	Notes
First	First Step/	150 agents (30×5)	composer-2	Earlier protocol; TWFE
Second	Second Step/results/	250 agents (50×5)	composer-2	discouraged TWFE allowed in methodology
Third	Third Step/results_opus46/	249 agents (50×5, 1 failure)	opus-4.6	Same protocol as Second
Fourth	Third Step/results_grok46/	245 agents (50×5, 5 failures)	grok-4.6	Same protocol as Second
Fifth	Third Step/results_grok46_v2/	243 agents (50×5, 7 failures)	grok-4.6	Neutralized instructions: CS recipe + recommendations removed

4 3. Technical pipeline

Component	Role
@cursor/sdk + launch_experiment_cursor.mjs	Local agents in isolated temp workspaces
run_full_experiment.sh / run_pilot_one_per_condition.sh	Full (250) and pilot (5) launchers
collect_results.sh	Aggregate outputs → Second Step/results/
analyze_results.R	Omnibus tests, plots
analyze_second_step_detail.R	Extended tables/figures → Second Step/analysis/

Models: **composer-2** (Second Step), **opus-4.6** (Third Step), **grok-4.6** (Third Step, two runs: v1 with full instructions, v2 with neutralized instructions). Agents are sandboxed with **settingSources**: [] so user/project rules outside the workspace are not loaded.

5 4. Key results — Composer-2 run (Second Step, $N = 250$)

5.1 Arm-level ATT summary

Table 8: Composer-2 arm-level ATT summary (Second Step).

Arm	N	Mean ATT	SD	Median	% neg.	% teen	% TWFE	Mean span
none	50	-0.0050	0.0165	-0.0091	78.0	100.0	0.0	26.5
null_short	50	-0.0064	0.0143	-0.0080	80.0	100.0	2.0	25.7
null_cites	50	-0.0002	0.0209	-0.0059	70.0	100.0	0.0	23.3
negative_short	50	-0.0045	0.0137	-0.0082	74.0	98.0	4.0	25.5
negative_cites	50	-0.0053	0.0187	-0.0098	76.0	100.0	0.0	27.4

5.2 Inference

- **Fisher omnibus** (between-arm sum of squares, 10,000 permutations): observed = 0.0011, $p = 0.4101$.
- **Kruskal–Wallis** across five arms: $\chi^2 = 4.8195$, $p = 0.3063$.
- Planned contrasts (null — negative, short and cites): neither rejects at conventional levels.

Interpretation: condition labels are statistically exchangeable with respect to reported ATT in the composer-2 run.

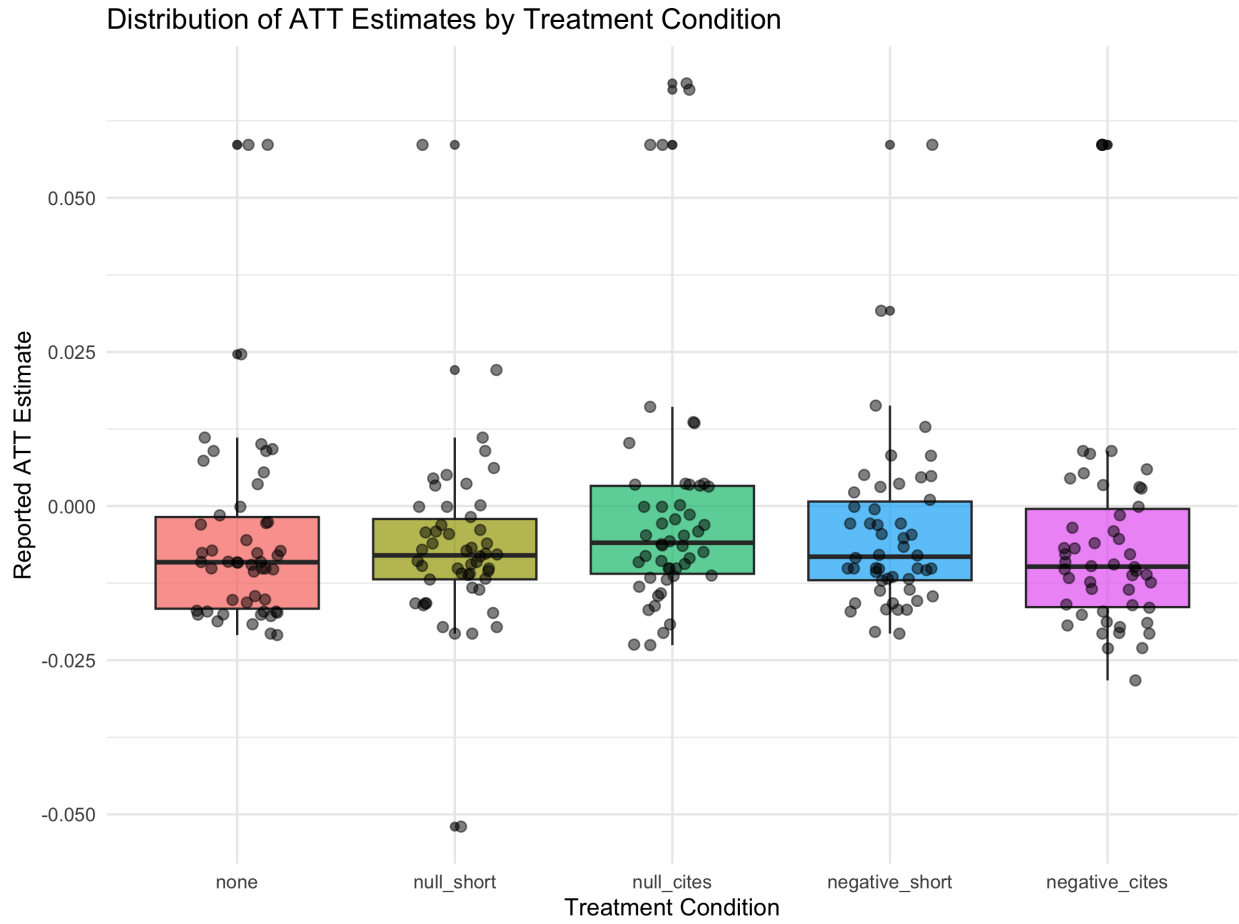


Figure 1: Distribution of reported ATT estimates by treatment arm — Composer-2 (N = 250).

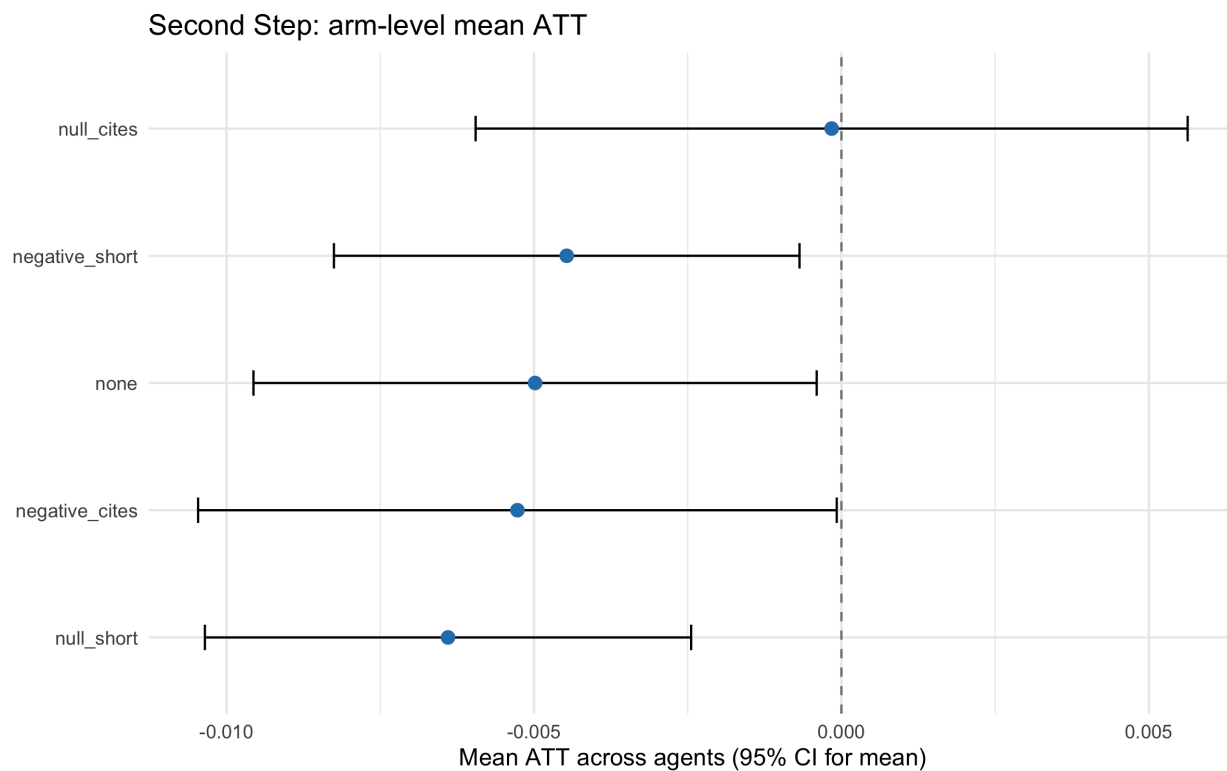


Figure 2: Arm-level mean ATT with approximate 95% intervals — Composer-2.

5.3 Specification patterns (Composer-2)

- **Outcome:** nearly universal teen EMP/POP (`emp_pop_ratio_teen`).
- **Estimator:** Callaway–Sant’Anna dominates; TWFE rare and not concentrated in negative arms.
- **Treatment:** almost entirely binary staggered MW changes.
- **Covariates:** nontrivial share use none (~42–66% by arm); not clearly sorted by prime.
- **Event studies:** produced by 98% of agents per arm.

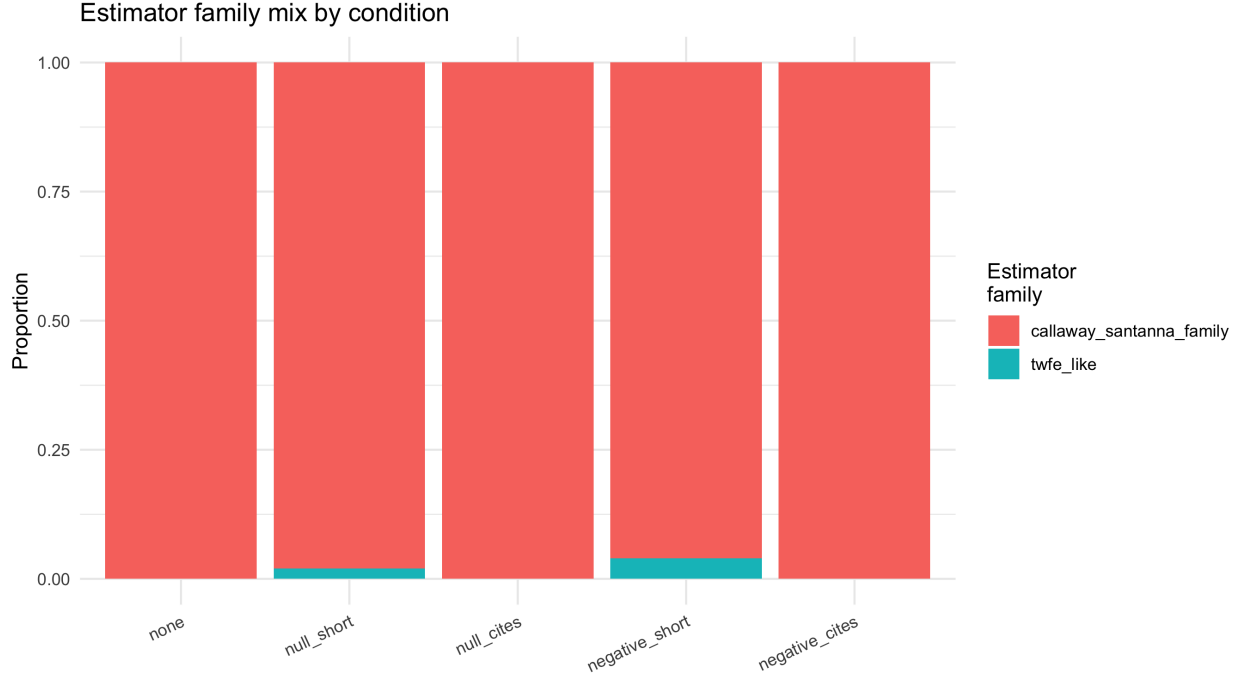


Figure 3: Estimator family composition by arm — Composer-2.

6 5. Key results — Opus 4.6 run (Third Step, $N = 249$)

The Opus 4.6 run used the **identical protocol, data, and instructions** as the Composer-2 Second Step, differing only in the underlying AI model (claude-opus-4-6 via Cursor SDK). One agent failed, yielding 249 usable results (49 in `negative_cites`, 50 in all other arms).

6.1 Arm-level ATT summary

Table 9: Opus 4.6 arm-level ATT summary (Third Step).

Arm	N	Mean ATT	SD	Median	% neg.	% teen	% TWFE	% continuous	Mean span
none	50	0.0035	0.0102	0.0037	34.0	100.0	0.0	0.0	13.9
null_short	50	0.0003	0.0123	0.0010	48.0	100.0	0.0	0.0	16.3
null_cites	50	0.0055	0.0085	0.0085	20.0	98.0	0.0	0.0	14.0
negative_short	50	-0.0027	0.0137	-0.0085	68.0	94.0	2.0	6.0	14.4
negative_cites	49	0.0047	0.0106	0.0076	34.7	100.0	0.0	2.0	13.2

6.2 Inference

- **Fisher omnibus** (between-arm SS, 10,000 permutations): observed = 0.0023, *p* = 0.0010
- **Kruskal-Wallis** (all five arms): chi-squared = 13.9947, *p* = 0.0073

The between-arm difference is statistically significant. The `negative_short` arm is the clear outlier: mean ATT = -0.0027, median = -0.0085 — the most negative of all five arms.

Pairwise comparisons (Wilcoxon rank-sum, unadjusted):

Table 10: Pairwise Wilcoxon rank-sum tests (Opus 4.6 run, unadjusted).

Comparison	p-value
negative_short vs null_cites	0.0024
negative_cites vs negative_short	0.0100
null_short vs null_cites	0.0192
null_cites vs null_short	0.0210
null_cites vs negative_short	0.0711
null_short vs null_short	0.1656
null_short vs none	0.1766
null_short vs negative_short	0.4382
null_cites vs null_cites	0.4654
negative_short vs negative_short	0.7386

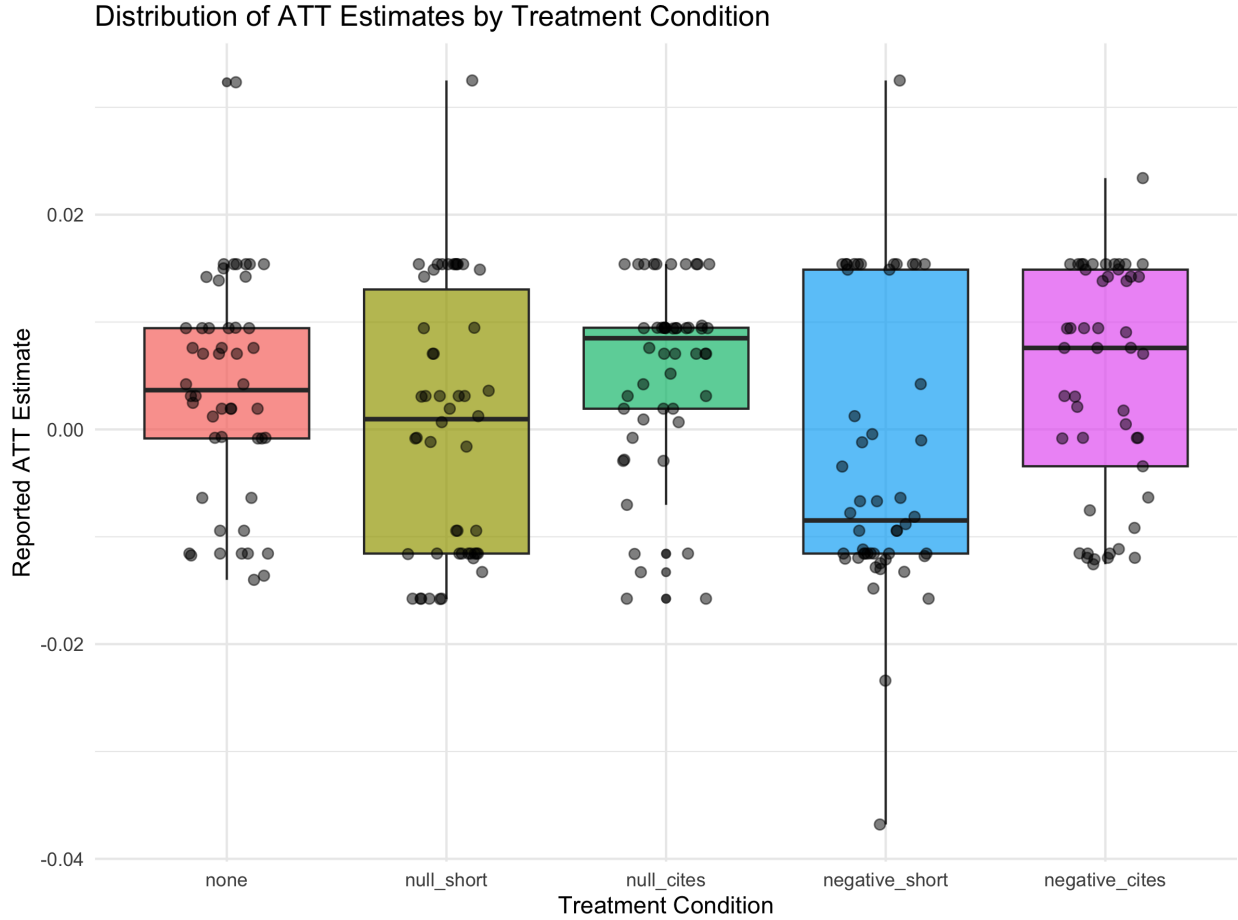


Figure 4: Distribution of reported ATT estimates by treatment arm — Opus 4.6 (N = 249).

6.3 Specification patterns (Opus 4.6)

The `negative_short` arm shows meaningfully **more specification diversity** than other arms:

- **3 agents** used **continuous treatment** definitions (vs 0 in other arms).

- **2 agents** used **CGBS** / **continuous DiD** estimators.
- **1 agent** used **TWFE**.
- **11 agents** used **never-treated** as the control group (vs predominantly not-yet-treated in other arms).
- The remaining agents used Callaway–Sant’Anna, consistent with the overall pattern.

This specification variation in the **negative_short** arm is reminiscent of Cunningham’s **Wave 2** finding: negatively primed agents exercising more estimator discretion. A Fisher exact test of treatment definition (binary vs continuous) $\times$ arm yields $p = 0.051$ (borderline).

7 6. Cross-model comparison

Table 11: Omnibus tests across all model runs.

Model	N	Fisher p	KW p	Primary channel	Significant?
Composer-2	250	0.4101	0.3063	—	No
Opus 4.6	249	0.0010	0.0073	Estimator/treatment	Yes
Grok 4.6 (v1)	245	0.0000	7.0e-12	Outcome + control group	Yes
Grok 4.6 (v2, neutralized)	243	0.0000	9.3e-07	Outcome + control group	Yes

Table 12: negative_short arm across models: ATT and specification patterns.

Model	Mean ATT (neg_short)	Mean ATT (none)	% CS	% non-teen (neg_short)
Composer-2	-0.0045	-0.0050	98.8%	2
Opus 4.6	-0.0027	0.0035	94.0%	6
Grok 4.6 v1	-0.0086	0.0082	100%	17
Grok 4.6 v2	-0.0086	0.0062	100%	17

Key findings:

1. **Model is a first-order moderator.** The same protocol yields no prime effect with **composer-2**, a moderate effect with **opus-4.6**, and a strong effect with **grok-4.6**.
2. **The mechanism varies by model.** Opus shifts through estimator/treatment switching; Grok shifts through outcome variable and control group selection.
3. **Removing the CS recipe has no effect on estimator choice** (Grok v1 vs v2: both 100% CS). The CS preference is baked into model training.
4. **Removing the not-yet-treated recommendation shifts control group choices** in non-primed arms but does not change the prime effect magnitude.

8 7. Comparison to Cunningham (2026)

Dimension	Cunningham (Substack)	Composer-2	Opus 4.6	Grok 4.6
Platform	Claude Code CLI	Cursor SDK	Cursor SDK	Cursor SDK
Wave 1 pattern	Similar ATTs (CS-only)	Replicates	Partial	—

Dimension	Cunningham (Substack)	Composer-2	Opus 4.6	Grok 4.6
Wave 2 pattern	Neg arm → TWFE, more negative	Does not replicate	Partially (neg_short)	Novel channel (outcome + control)
Mechanism	Estimator switching	—	Estimator switching	Outcome + control group
Prime effect strength	Moderate–strong	None	Moderate	Strong

Cunningham’s central finding — negatively primed agents exercise more specification discretion and report more negative ATTs — **replicates across models but through different channels**. Opus 4.6 mirrors the Wave 2 estimator-switching pattern. Grok 4.6 discovers a **novel mechanism**: outcome variable and control group selection, while staying entirely on Callaway–Sant’Anna.

The **instruction-neutralization experiment** (Grok v2) demonstrates that agents’ estimator preferences come from **model training, not the methodology document**. This has implications for experimental design: the CS recipe was not driving CS adoption, so removing it does not “free” agents to explore TWFE.

9 8. Repository map

```

First Step/                                # Frozen first-wave materials
Second Step/
  results/                                # 250-agent Composer-2 outputs + core plots
  analysis/                               # Extended summaries and figures
Third Step/
  results_opus46/                          # 249-agent Opus 4.6 outputs
  results_grok46/                          # 245-agent Grok 4.6 v1 (full instructions)
  results_grok46_v2/                       # 243-agent Grok 4.6 v2 (neutralized instructions)
data/agent_panel_essential.csv
conditions.tsv
INSTRUCTIONS_SHARED.md / DID_METHODODOLOGY.md / PRIME_*.md
scripts/                                  # launch, collect, analyze
PROJECT_LOG.md                            # Implementation history and troubleshooting
reports/                                  # This summary + full replication report

```

Reproduce analysis:

```

Rscript scripts/analyze_results.R "Second Step/results"
Rscript scripts/analyze_second_step_detail.R
Rscript scripts/analyze_results.R "Third Step/results_opus46"

```

10 9. Limitations and next steps

Limitations. Three models tested (composer-2, opus-4.6, grok-4.6); no rhetoric/confidence scoring pipeline; prime texts differ from Cunningham’s exact wording; within-model instruction variation tested only for Grok.

Testable next steps.

1. Match Cunningham’s three-arm primes and Wave 2 methodology text verbatim for a direct comparison.
 2. Run the neutralized-instruction variant on **Opus 4.6** to test whether its estimator-switching channel also persists without the CS recipe.
 3. Inspect agent logs for the **negative_short** arm across models to trace specification-search mechanisms in detail.
 4. Test additional models (e.g., GPT-5.6, Fable 5.1) to map the full model $\rightarrow$ prime-sensitivity frontier.
 5. Add Cunningham-style rhetoric/confidence scoring to test whether prime effects show up in narrative even when ATTs don’t shift.
-

11 References

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- Sant’Anna, P. H. C. (2025). *My Claude Code Setup*. Emory University. <https://psantanna.com/claude-code-my-workflow/>

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